

Matrix – Determinant – System of Linear Equations

System of Linear Equations

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System of Linear Equations

- 1 **System of Linear Equations**
- 2 Applications of Linear Systems
- 3 Exercises



Definition (System of Linear Equations)

A **system of linear equations** (or a **linear system**) is a collection of one or more linear equations involving the same variables. A linear system of m equations in n variables $x_1, x_2, \dots, x_n$ has the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}, \left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right).$$

The matrix representation of this system is compactly written as

$$AX = b, \quad A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{pmatrix},$$

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Definition

A system of linear equations may have

- 1 no solution
- 2 exactly one solution
- 3 infinitely many solutions.

Definition

A system of linear equations is

- 1 **consistent** if it has at least one solution (either exactly one solution or infinitely many solutions).
- 2 **inconsistent** if it has no solution.



Example 1.

Solve the system
$$\begin{cases} x_1 + 2x_2 + x_3 = 2 \\ 2x_1 + x_2 + 2x_3 = 3 \\ x_1 + 5x_2 + x_3 = 3, \end{cases}$$



Example 1.

Solve the system
$$\begin{cases} x_1 + 2x_2 + x_3 = 2 \\ 2x_1 + x_2 + 2x_3 = 3 \\ x_1 + 5x_2 + x_3 = 3, \end{cases}$$

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 5 & 1 \end{pmatrix}, X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, b = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} : AX = b$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 5 & 1 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & -1 \\ 0 & 3 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\text{rank}(A) = \text{rank}(A|b) = 2 < n = 3$$

$\Rightarrow$ the system has infinitely many solution, and $n - \text{rank}(A) = 1$ free variable x_3 .



Example 1.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 5 & 1 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & -1 \\ 0 & 3 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\text{rank}(A) = \text{rank}(A|b) = 2 < n = 3$$

$\Rightarrow$ the system has infinitely many solutions, and $n - \text{rank}(A) = 1$ free variable x_3 .

Choose $x_3 = t$. From (2) $\iff x_2 = \frac{1}{3}$. From (1): $x_1 + \frac{2}{3} + t = 2 \iff x_1 = \frac{4}{3} - t$.

Therefore $\begin{cases} x_1 = \frac{4}{3} - t \\ x_2 = \frac{1}{3} \\ x_3 = t, \end{cases} \quad t \in \mathbb{R}.$



Definition (Basic and Free Variables)

The variables corresponding to pivot columns in the matrix are called **basic variables**.
The other variables are called **free variables**.



Example 2.

Solve the system
$$\begin{cases} x_1 + x_2 - x_3 + x_4 = 1 \\ 2x_1 + 2x_2 + 2x_3 + 2x_4 = 1 \\ 3x_1 + 3x_2 + 5x_3 + 3x_4 = 3, \end{cases}$$



Example 2.

Solve the system
$$\begin{cases} x_1 + x_2 - x_3 + x_4 = 1 \\ 2x_1 + 2x_2 + 2x_3 + 2x_4 = 1 \\ 3x_1 + 3x_2 + 5x_3 + 3x_4 = 3, \end{cases}$$

$$A = \begin{pmatrix} 1 & 1 & -1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 5 & 3 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}, X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 1 \\ 3 & 3 & 5 & 1 & 3 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 1 \\ 0 & 0 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 \end{array} \right]$$

Because $\text{rank}(A) = 2 < \text{rank}(A|b) = 3$, the system is inconsistent.



Example 3.

Solve the system
$$\begin{cases} x_1 + x_2 + x_3 = 5 \\ 2x_1 + 3x_2 + 2x_3 = 8 \\ 3x_1 + 4x_2 - x_3 = -3 \\ 3x_1 + 4x_2 - 5x_3 = -19 \end{cases}$$



Example 3.

Solve the system
$$\begin{cases} x_1 + x_2 + x_3 = 5 \\ 2x_1 + 3x_2 + 2x_3 = 8 \\ 3x_1 + 4x_2 - x_3 = -3 \\ 3x_1 + 4x_2 - 5x_3 = -19 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 2 & 3 & 2 & 8 \\ 3 & 4 & -1 & -3 \\ 3 & 4 & -5 & -19 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & -4 & -16 \\ 0 & 0 & 0 & 0 \end{array} \right], \text{rank}(A) = \text{rank}(A|b) = n$$

$\Rightarrow$ the system has exactly one solution.

From (3): $x_3 = 4$

From (2): $x_2 = -2$

From (1): $x_1 = 5 - x_2 - x_3 = 3$.

The solution is
$$\begin{cases} x_1 = 3 \\ x_2 = -2, x_3 = 4 \end{cases}$$



Kronecker Capelli Theorem

Given a linear system $Ax = b$, $A \in M_{m \times n}$.

- $\text{rank}(A) < \text{rank}(A|b)$: the system is inconsistent.
- $\text{rank}(A) = \text{rank}(A|b) = r$: the system is consistent.
 $\longrightarrow \begin{cases} r = n : \text{the system has exactly one solution,} \\ r < n : \text{the system has infinitely many solution.} \end{cases}$



Kronecker Capelli Theorem

Given a linear system $Ax = b$, $A \in M_{m \times n}$.

- $\text{rank}(A) < \text{rank}(A|b)$: the system is inconsistent.
- $\text{rank}(A) = \text{rank}(A|b) = r$: the system is consistent.
 $\longrightarrow \begin{cases} r = n : \text{the system has exactly one solution,} \\ r < n : \text{the system has infinitely many solution.} \end{cases}$

Another Version

Given a linear system $Ax = b$, $A \in M_{m \times n}$.

- The system is consistent iff the rightmost column of the augmented matrix is not a pivot column.
- The system is consistent, then the solution set contains either:
 $\longrightarrow \begin{cases} \text{a unique solution, when there are no free variables, or,} \\ \text{infinitely many solutions, when there is at least one free variable.} \end{cases}$

Example 4.

Solve the system
$$\begin{cases} x_1 + x_2 - x_3 + 2x_4 = 1 \\ 2x_1 + 3x_2 - 3x_3 + 3x_4 = 3 \\ 3x_1 + 2x_2 - 5x_3 + 7x_4 = 5. \end{cases}$$



Example 4.

Solve the system
$$\begin{cases} x_1 + x_2 - x_3 + 2x_4 = 1 \\ 2x_1 + 3x_2 - 3x_3 + 3x_4 = 3 \\ 3x_1 + 2x_2 - 5x_3 + 7x_4 = 5. \end{cases}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 1 \\ 2 & 3 & -3 & 3 & 3 \\ 3 & 2 & -5 & 7 & 5 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & -3 & 0 & 3 \end{array} \right]$$

$\text{rank}(A) = \text{rank}(A|b) = 3 < n = 4$: the system has infinitely many solutions.



Example 4.

Solve the system
$$\begin{cases} x_1 + x_2 - x_3 + 2x_4 = 1 \\ 2x_1 + 3x_2 - 3x_3 + 3x_4 = 3 \\ 3x_1 + 2x_2 - 5x_3 + 7x_4 = 5. \end{cases}$$

$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 1 \\ 2 & 3 & -3 & 3 & 3 \\ 3 & 2 & -5 & 7 & 5 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 1 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & -3 & 0 & 3 \end{array} \right]$$

$\text{rank}(A) = \text{rank}(A|b) = 3 < n = 4$: the system has infinitely many solutions.

Let $x_4 = t$. From eq.(3): $x_3 = -1$

eq.(2): $x_2 + 1 - t = 1 \iff x_2 = t$.

eq.(1): $x_1 + t + 1 + 2t = 1 \iff x_1 = -3t$.

The general solution of the system is $(x_1, x_2, x_3, x_4) = (-3t, t, -1, t), t \in \mathbb{R}$.



Example 5.

Solve the system
$$\begin{cases} x_1 + x_2 + x_3 + 2x_4 &= 1 \\ 2x_1 + 2x_2 + 3x_3 + 5x_4 &= 6 \\ x_1 + x_2 &+ x_4 &= -3 \end{cases}$$



Example 5.

Solve the system
$$\begin{cases} x_1 + x_2 + x_3 + 2x_4 &= 1 \\ 2x_1 + 2x_2 + 3x_3 + 5x_4 &= 6 \\ x_1 + x_2 &+ x_4 &= -3 \end{cases}$$

$$(A|b) = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 2 & 1 \\ 2 & 2 & 3 & 5 & 6 \\ 1 & 1 & 0 & 1 & -3 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

$\text{rank}(A) = \text{rank}(A|b) = 2 < n = 4$: the system has infinitely many solutions.

Choose $x_2 = \alpha, x_4 = \beta$.

From eq.(2): $x_3 = 4 - x_4 = 4 - \beta$.

From eq.(1): $x_1 = 1 - x_2 - x_3 - 2x_4 = 1 - \alpha - 4 + \beta - 2\beta = -3 - \alpha - \beta$

The general solution of the system is

$$(x_1; x_2; x_3; x_4) = (-3 - \alpha - \beta; \alpha; 4 - \beta; \beta), \forall \alpha, \beta \in R.$$



Example 6.

Solve the system
$$\begin{cases} x_1 + 2x_2 - x_3 + 3x_4 = -12 \\ 2x_1 + x_2 + 3x_3 - 2x_4 = 13 \\ 4x_1 + 5x_2 + x_3 + 4x_4 = -11 \\ x_1 + 5x_2 - 6x_3 + 11x_4 = -49 \end{cases}$$



Example 6.

$$\text{Solve the system } \begin{cases} x_1 + 2x_2 - x_3 + 3x_4 = -12 \\ 2x_1 + x_2 + 3x_3 - 2x_4 = 13 \\ 4x_1 + 5x_2 + x_3 + 4x_4 = -11 \\ x_1 + 5x_2 - 6x_3 + 11x_4 = -49 \end{cases} \rightarrow \left[\begin{array}{cccc|c} 1 & 2 & -1 & 3 & -12 \\ 0 & -3 & 5 & -8 & 37 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

We have $\text{rank}(A) = \text{rank}(A|b) = 2 < n = 4 \Rightarrow$ the system has infinitely many solutions, with $n - \text{rank}(A) = 2$ free variables.

Choose $x_3 = a; x_4 = b$. From eq.(2) : $-3x_2 + 5a - 8b = 37 \iff x_2 = \frac{-37}{3} + \frac{5}{3}a - \frac{8}{3}b$

From eq.(1) : $x_1 - \frac{74}{3} + \frac{10}{3}a - \frac{16}{3}b - a + 3b = 12 \iff x_1 = \frac{38}{3} - \frac{7}{3}a + \frac{7}{3}b$.

Therefore, $(x_1; x_2; x_3; x_4) = \left(\frac{38}{3} - \frac{7}{3}a + \frac{7}{3}b; \frac{37}{3} - \frac{5}{3}a + \frac{8}{3}b; a; b \right), a, b \in \mathbb{R}$.



Theorem. Cramer's rule

Let A be an invertible $n \times n$ matrix. For any b in $\mathbb{R}^n$, the unique solution x of $Ax = b$ has entries given by

$$x_i = \frac{\det(A_i(b))}{\det(A)}, \quad i = 1, 2, \dots, n$$

where $A_i(b)$ be the matrix obtained from A by replacing column i by the vector b , i.e.

$$A_i(b) = [a_1 \cdots \underset{\substack{\uparrow \\ i}}{b} \cdots a_n]$$



Example 7.

Use Cramer's rule to solve the system
$$\begin{cases} 3x_1 - 2x_2 = 6 \\ -5x_1 + 4x_2 = 8 \end{cases}$$



Example 7.

Use Cramer's rule to solve the system
$$\begin{cases} 3x_1 - 2x_2 = 6 \\ -5x_1 + 4x_2 = 8 \end{cases}$$

$$A = \begin{pmatrix} 3 & -2 \\ -5 & 4 \end{pmatrix}, \quad A_1(b) = \begin{pmatrix} 6 & -2 \\ 8 & 4 \end{pmatrix}, \quad A_2(b) = \begin{pmatrix} 3 & 6 \\ -5 & 8 \end{pmatrix}.$$

Since $\det A = 2$, the system has a unique solution. By Cramer's rule,

$$x_1 = \frac{\det(A_1(b))}{\det(A)} = \frac{24 + 16}{2} = 20$$

$$x_2 = \frac{\det(A_2(b))}{\det(A)} = \frac{24 + 30}{2} = 27$$



Homogeneous systems

A system is called homogeneous if all its constant terms $b_1, \dots, b_m$ are zero, i.e. $AX = 0, A \in M_n, X \in \mathbb{R}_{m \times n}$. The system always has zero solution $X = 0$ (trivial solution).

$\left[\begin{array}{l} \text{rank}(A) = n : \text{ the system always possesses a unique solution } X=0 \\ \text{rank}(A) < n : \text{ the system has infinitely many solution(nontrivial solution).} \end{array} \right.$



Example 8.

Find m such that $\begin{cases} x_1 + x_2 + mx_3 = 0 \\ 2x_1 + x_2 - x_3 = 0 \\ x_1 + 2x_2 + mx_3 = 0 \end{cases}$ has only the trivial solution.



Example 8.

Find m such that $\begin{cases} x_1 + x_2 + mx_3 = 0 \\ 2x_1 + x_2 - x_3 = 0 \\ x_1 + 2x_2 + mx_3 = 0 \end{cases}$ has only the trivial solution.

The homogenous system has only the trivial solution iff

$$\begin{vmatrix} 1 & 1 & m \\ 2 & 1 & -1 \\ 1 & 2 & m \end{vmatrix} \neq 0 \iff 2m + 1 \neq 0 \iff m \neq -\frac{1}{2}.$$



Example 9.

Find m such that
$$\begin{cases} mx_1 + x_2 + x_3 + x_4 = 0 \\ x_1 + mx_2 + x_3 + x_4 = 0 \\ x_1 + x_2 + mx_3 + x_4 = 0 \\ x_1 + x_2 + x_3 + mx_4 = 0 \end{cases}$$
 has nontrivial solution.



Example 9.

Find m such that $\begin{cases} mx_1 + x_2 + x_3 + x_4 = 0 \\ x_1 + mx_2 + x_3 + x_4 = 0 \\ x_1 + x_2 + mx_3 + x_4 = 0 \\ x_1 + x_2 + x_3 + mx_4 = 0 \end{cases}$ has nontrivial solution.

The homogeneous system has nontrivial solution (infinitely many solutions) iff $\text{rank}(A) < n \iff \det(A) = 0$.

$$|A| = \begin{vmatrix} m & 1 & 1 & 1 \\ 1 & m & 1 & 1 \\ 1 & 1 & m & 1 \\ 1 & 1 & 1 & m \end{vmatrix} = (m+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & m & 1 & 1 \\ 1 & 1 & m & 1 \\ 1 & 1 & 1 & m \end{vmatrix} = (m+3)(m-1)^3.$$

Thus $\begin{cases} m = -3 \\ m = 1 \end{cases}.$



Example 9.

Find m such that
$$\begin{cases} x_1 + 2x_2 - x_3 + mx_4 = 2 \\ x_1 + mx_2 - x_3 + x_4 = 1 \\ x_1 + 2x_2 + (m+1)x_4 = 3 \end{cases}$$
 has exactly one solution.



Example 9.

Find m such that $\begin{cases} x_1 + 2x_2 - x_3 + mx_4 = 2 \\ x_1 + mx_2 - x_3 + x_4 = 1 \\ x_1 + 2x_2 + (m+1)x_4 = 3 \end{cases}$ has exactly one solution.

The system has exactly one solution $\iff \text{rank}(A) = \text{rank}(A|b) = n$.

However, $A \in M_{3 \times 4}$ so $\text{rank}(A) \leq 3 < n = 4$.

Hence, no value of m exists such that the system has a unique solution.



Example 10.

Find m such that
$$\begin{cases} x_1 + x_2 - 2x_3 = 1 \\ 2x_1 + 3x_2 - 3x_3 = 5 \\ 3x_1 + mx_2 - 7x_3 = 8. \end{cases}$$
 has no solution.



Example 10.

Find m such that $\begin{cases} x_1 + x_2 - 2x_3 = 1 \\ 2x_1 + 3x_2 - 3x_3 = 5 \\ 3x_1 + mx_2 - 7x_3 = 8. \end{cases}$ has no solution.

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 2 & 3 & -3 & 5 \\ 3 & m & -7 & 8 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & m-2 & 3m-14 \end{array} \right]$$

The system has no solution iff

$$\text{rank}(A) < \text{rank}(A|b) \iff \begin{cases} m-2 = 0 \\ 3m-14 \neq 0 \end{cases} \iff m = 2.$$



Example 11.

Find m such that $\begin{cases} x_1 + 2x_2 + x_3 = 5 \\ 2x_1 + x_2 + x_3 = 2m - 1 \\ x_1 + 5x_2 + mx_3 = 12 \end{cases}$ has a solution.



Example 11.

Find m such that $\begin{cases} x_1 + 2x_2 + x_3 = 5 \\ 2x_1 + x_2 + x_3 = 2m - 1 \\ x_1 + 5x_2 + mx_3 = 12 \end{cases}$ has a solution.

$$(A|b) = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 2 & 1 & 1 & 2m-1 \\ 1 & 5 & m & 12 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & -3 & -1 & 2m-11 \\ 0 & 0 & m-2 & 2m-4 \end{array} \right].$$

The system has a solution $\iff r(A) = r(A|b) \iff \forall m \in \mathbb{R}$.



Example 12.

Find m such that
$$\begin{cases} x_1 + x_2 + x_3 &= 1 \\ x_1 + (m - 1)x_2 &= -1 \\ x_1 + 2mx_2 + (1 + m)x_3 &= -2 - m \end{cases}$$
 has infinitely many solutions.



Example 12.

Find m such that $\begin{cases} x_1 + x_2 + x_3 &= 1 \\ x_1 + (m-1)x_2 &= -1 \\ x_1 + 2mx_2 + (1+m)x_3 &= -2-m \end{cases}$ has infinitely many solutions.

$$(A|b) = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & m-1 & 0 & -1 \\ 1 & 2m & 1+m & -2-m \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & m-2 & -1 & -2 \\ 0 & 2m-1 & m & -3-m \end{array} \right]$$

$$\xrightarrow{2r_2-r_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -3 & -2-m & m \\ 0 & 2m-1 & m & -3-m \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -3 & -2-m & -1+m \\ 0 & 0 & 2-2m^2 & 2m^2-6m-8 \end{array} \right]$$

The system has infinitely many solution $\iff \begin{cases} 2-2m^2=0 \\ 2m^2-6m-8=0 \end{cases} \iff m = -1$



Example 13.

Find m such that
$$\begin{cases} x_1 - x_2 + 2x_3 & = 1 \\ x_1 - x_2 + (m+1)x_3 & = m+2 \\ -3x_1 + 3x_2 + (2m-9)x_3 & = m \end{cases}$$
 has no solution.



Example 13.

Find m such that $\begin{cases} x_1 - x_2 + 2x_3 &= 1 \\ x_1 - x_2 + (m+1)x_3 &= m+2 \\ -3x_1 + 3x_2 + (2m-9)x_3 &= m \end{cases}$ has no solution.

$$\begin{aligned} (A|b) &= \left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 1 & -1 & m+1 & m+2 \\ -3 & 3 & 2m-9 & m \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 0 & m-1 & m+1 \\ 0 & 0 & 2m-3 & m+3 \end{array} \right] \\ &\xrightarrow{2r_2-r_3} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & m-1 \\ 0 & 0 & 2m-3 & m+3 \end{array} \right] \xrightarrow{r_3-(2m-3)r_2} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & m-1 \\ 0 & 0 & 0 & -2m^2+6m \end{array} \right] \end{aligned}$$

The system has no solution

$$\iff r(A) < r(A|b) \iff -2m^2 + 6m \neq 0 \iff \begin{cases} m \neq 0 \\ m \neq 3 \end{cases}.$$



Example 14.

Determine the number of solutions of the system
$$\begin{cases} mx_1 + x_2 + x_3 = 1 \\ x_1 + mx_2 + x_3 = m \\ x_1 + x_2 + mx_3 = m^2 \end{cases}.$$



Applications of Linear Systems

- 1 System of Linear Equations
- 2 Applications of Linear Systems**
- 3 Exercises



Direct current circuit problem

- Electric current: denoted by I , with unit A (ampere).
- Resistance: denoted by R , with unit Ω (ohm).
- Voltage (potential difference): denoted by U , with unit V (volt).
- Electric current flows from higher potential to lower potential.
- Ohm's law: $V = RI$.

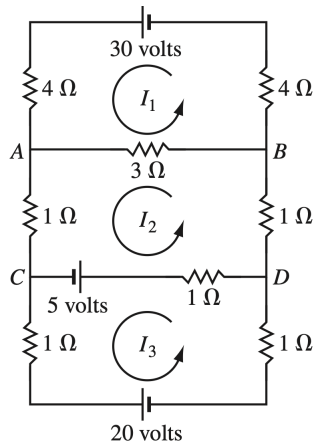
Kirchoff's Laws

- **KCL – Kirchhoff Current Law.** The total current entering a junction equals the total current leaving the junction.
- **KVL – Kirchhoff Voltage Law.** The algebraic sum of all voltages around any closed loop in a circuit is zero.



Example 15.

Determine the loop currents in the network in the figure.



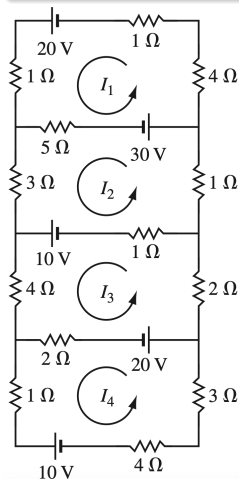
Kirchhoff's voltage law implies that

$$\begin{cases} \text{Loop 1: } 11I_1 - 3I_2 = 30, \\ \text{Loop 2: } -3I_1 + 6I_2 - I_3 = 5, \\ \text{Loop 3: } -I_2 + 3I_3 = -25 \end{cases}$$

$$I_1 = 3, I_2 = 1, I_3 = -8(\text{Amps}).$$

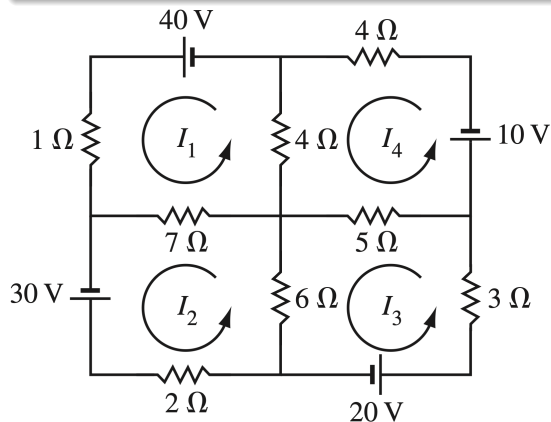
Example 16.

Determine the loop currents in the network in the figure.



Example 17.

Determine the loop currents in the network in the figure.



Leontief Input–Output Model

Assume an economy has three sectors: industry, agriculture, and services (values are measured in USD).

Intermediate demand x_{ij} is the value of goods that sector i supplies to sector j as inputs for production.

Final demand b_i is the value of goods from sector i required for labour, consumption, exports, and other non-intermediate uses.

Total output x_i is the total value of goods produced in sector i , equal to the sum of intermediate demand and final demand.

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Final demand b_i is the value of goods from sector i required for labour, consumption, exports, and other non-intermediate uses.

Total output x_i is the total value of goods produced in sector i , equal to the sum of intermediate demand and final demand.

$$\left\{ \begin{array}{l} x_1 = x_{11} + x_{12} + x_{13} + b_1 \\ x_2 = x_{21} + x_{22} + x_{23} + b_2 \\ x_3 = x_{31} + x_{32} + x_{33} + b_3 \end{array} \right. \iff \left\{ \begin{array}{l} x_1 = \frac{x_{11}}{x_1} x_1 + \frac{x_{12}}{x_2} x_2 + \frac{x_{13}}{x_3} x_3 + b_1 \\ x_2 = \frac{x_{21}}{x_1} x_1 + \frac{x_{22}}{x_2} x_2 + \frac{x_{23}}{x_3} x_3 + b_2 \\ x_3 = \frac{x_{31}}{x_1} x_1 + \frac{x_{32}}{x_2} x_2 + \frac{x_{33}}{x_3} x_3 + b_3 \end{array} \right.$$

I/O model

Let $a_{ij} = \frac{x_{ij}}{x_j}$: be the direct input coefficient (based on the total output of sector j)..

→ to produce 1 (USD) of output in sector j , sector j requires a_{ij} USD worth of goods from sector i .

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$$\begin{cases} x_1 = a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + b_1 \\ x_2 = a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + b_2 \\ x_3 = a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + b_3 \end{cases} \iff \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

$$X = AX + b, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

The matrix A is called the direct input coefficient matrix (also known as the technical matrix or input matrix).

Example 18.

The balance model for the three sectors — industry, agriculture, and services — of a country has the following direct input-coefficient matrix: $A = \begin{pmatrix} 0.3 & 0.05 & 0.15 \\ 0.1 & 0.25 & 0.2 \\ 0.1 & 0.1 & 0.2 \end{pmatrix}$

a/ State the meaning of each numerical value in the matrix..

b/ In 2021, the industry, agriculture, and services sectors produced 2000; 5000; and 3000 million USD respectively. Compute the final demand and explain its meaning.



Example 18.

The balance model for the three sectors — industry, agriculture, and services — of a country has the following direct input-coefficient matrix: $A = \begin{pmatrix} 0.3 & 0.05 & 0.15 \\ 0.1 & 0.25 & 0.2 \\ 0.1 & 0.1 & 0.2 \end{pmatrix}$

a/ State the meaning of each numerical value in the matrix..

b/ In 2021, the industry, agriculture, and services sectors produced 2000; 5000; and 3000 million USD respectively. Compute the final demand and explain its meaning.

We have $X = \begin{pmatrix} 2000 \\ 5000 \\ 3000 \end{pmatrix}$. Final demand

$$b = (I - A)X = \begin{pmatrix} 0.7 & -0.05 & -0.15 \\ -0.1 & 0.75 & -0.2 \\ -0.1 & -0.1 & 0.8 \end{pmatrix} \begin{pmatrix} 2000 \\ 5000 \\ 3000 \end{pmatrix} = \begin{pmatrix} 700 \\ 2950 \\ 1700 \end{pmatrix}.$$



Example 19.

If possible, find some combination of nonfat milk, soy flour, and whey to provide the exact amounts of protein, carbohydrate, and fat supplied by the diet in one day.

Amounts (g) Supplied per 100 g of Ingredient				Amounts (g) Supplied by Cambridge Diet in One Day
Nutrient	Nonfat milk	Soy flour	Whey	
Protein	36	51	13	33
Carbohydrate	52	34	74	45
Fat	0	7	1.1	3

Let x_1 , x_2 , and x_3 , respectively, denote the number of units (100g) of these foodstuffs.

Thus, $x_1 a_1 + x_2 a_2 + x_3 a_3 = b$, where a_i , $i = 1, 2, 3$ is the i -th column in the table.

$$\begin{pmatrix} 36 & 51 & 13 & 33 \\ 52 & 34 & 74 & 45 \\ 0 & 7 & 1.1 & 3 \end{pmatrix} \sim \dots \sim \begin{pmatrix} 1 & 0 & 0 & 0.277 \\ 0 & 1 & 0 & 0.392 \\ 0 & 0 & 1 & 0.233 \end{pmatrix} \quad \text{The diet requires 0.277 units of}$$

nonfat milk, 0.392 units of soy flour, and 0.233 units of whey in order to provide the desired amounts of protein, carbohydrate, and fat.

Example 20.

Assume a company has four raw materials for steel production: A, B, C, and D. Their compositions (in percent by mass) are:

- A: 5% Al, 3% Si, 4% C, 88% Fe
- B: 7% Al, 6% Si, 5% C, 82% Fe
- C: 2% Al, 1% Si, 3% C, 94% Fe
- D: 1% Al, 2% Si, 1% C, 96% Fe

Let (x_1, x_2, x_3, x_4) be the mass percentage proportions of materials (A, B, C, D) that must be mixed to produce a final alloy containing (by mass): Al: 5%, Si: 4%, C: 4%, Fe: 87%. Find the mass percentage of each material.



Example 21.

There are three solutions X , Y , and Z of the same solute, with concentrations of 20%, 25%, and 40% respectively. These three solutions are mixed together to form a mixture with a concentration of 30%.

- a. Find the percentage of the volume of each solution in the mixture.
- b. Given that the volume of X is twice the volume of Y , and the total volume of the mixture is 50 litres, calculate the volume of each solution used.



Exercise 1. Solve the system

$$1/ \begin{cases} x_1 + 2x_2 + x_3 + 2x_4 = 0 \\ 2x_1 + 4x_2 + x_3 + 3x_4 = 0 \\ 3x_1 + 6x_2 + x_3 + 4x_4 = 0 \end{cases}$$

$$2/ \left[\begin{array}{ccc|c} 1 & 5 & 2 & 1 \\ -1 & -4 & 1 & 6 \\ 1 & 3 & -3 & -9 \end{array} \right]$$

$$3/ \left[\begin{array}{ccc|c} 1 & 5 & 2 & -6 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 5 & 0 \end{array} \right].$$

$$4/ \left[\begin{array}{ccc|c} 0 & 1 & 1 & 3 \\ 3 & 5 & 9 & -2 \\ 1 & 2 & 3 & 3 \end{array} \right]$$

$$5/ \left[\begin{array}{ccccc|c} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{array} \right]$$

$$6/ \left[\begin{array}{cccc|c} 1 & 1 & 1 & -1 & 2 \\ 2 & 1 & 3 & 0 & 1 \\ 3 & 4 & 2 & -2 & 5 \\ 2 & 3 & 1 & -1 & 3 \end{array} \right]$$



Exercise 1. Solve the system.

$$\begin{aligned} 1/ \quad & \begin{cases} x_1 + 2x_2 - x_3 = 12 \\ 2x_1 + 3x_2 - 3x_3 = 4 \\ 3x_1 + 2x_2 + 5x_3 = -8 \end{cases} \\ 2/ \quad & \begin{cases} x_1 + x_2 - x_3 + 2x_4 = 0 \\ 2x_1 + 3x_2 - 3x_3 + 3x_4 = 0 \\ 3x_1 + 5x_2 - 5x_3 + 4x_4 = 0. \end{cases} \end{aligned}$$



Exercise 2.

1/ Find m such that $\begin{cases} x_1 + x_2 + mx_3 = 0 \\ 2x_1 + x_2 - x_3 = 0 \\ x_1 + 2x_2 + mx_3 = 0 \end{cases}$ has a unique solution $X = 0$.

2/ Find m such that the system $\begin{cases} mx_1 + 2x_2 + 2x_3 + 2x_4 = 0 \\ 2x_1 + mx_2 + 2x_3 + 2x_4 = 0 \\ 2x_1 + 2x_2 + mx_3 + 2x_4 = 0 \\ 2x_1 + 2x_2 + 2x_3 + mx_4 = 0. \end{cases}$ has infinitely many solutions.



Exercise 2

3/ Find m such that the system $\begin{cases} x_1 + 2x_2 - x_3 + mx_4 = 2 \\ x_1 + mx_2 - x_3 + x_4 = 1 \\ x_1 + 2x_2 + (m+1)x_4 = 3 \end{cases}$ has exactly one solution.

4/ Find $m \in \mathbb{R}$ such that the system $\begin{cases} x_1 + x_2 + 2x_3 - x_4 = 0 \\ x_1 + 3x_2 + mx_3 + 2x_4 = 0 \\ mx_1 - x_2 + 3x_3 - 2x_4 = 0 \end{cases}$ has infinitely many solutions.



Exercise 2.

5/ Find all values of m such that the system $\begin{cases} x_1 + x_2 - 2x_3 = 1 \\ 2x_1 + 3x_2 - 3x_3 = 5 \\ 3x_1 + mx_2 - 7x_3 = 8. \end{cases}$ has no solution.

6/ Find all values of $m \in \mathbb{R}$ such that the system $\begin{cases} x_1 + 2x_2 + x_3 = 5 \\ 2x_1 + x_2 + x_3 = 2m - 1 \\ x_1 + 5x_2 + mx_3 = 12 \end{cases}$ has a solution.



Exercise 2

7/ Find m such that the system $\begin{cases} x_1 + x_2 + x_3 & = 1 \\ x_1 + (m - 1)x_2 & = -1 \\ x_1 + 2mx_2 + (1 + m)x_3 & = -2 - m \end{cases}$ has a solution

8/ Find m such that the system $\begin{cases} x_1 - x_2 + 2x_3 & = 1 \\ x_1 - x_2 + (m + 1)x_3 & = m + 2 \\ -3x_1 + 3x_2 + (2m - 9)x_3 & = m \end{cases}$ has no solution.



Exercise 3.

Discuss **the number of solutions** of the system of equations

$$\begin{cases} mx_1 + x_2 + x_3 = 1 \\ x_1 + mx_2 + x_3 = m \\ x_1 + x_2 + mx_3 = m^2 \end{cases}$$



THANK YOU FOR ATTENTION

